
Table of Contents

Preface.....	xi
1. Introduction.....	1
What Is Linear Algebra and Why Learn It?	1
About This Book	2
Prerequisites	2
Math	3
Attitude	3
Coding	3
Mathematical Proofs Versus Intuition from Coding	4
Code, Printed in the Book and Downloadable Online	5
Code Exercises	5
How to Use This Book (for Teachers and Self Learners)	6
2. Vectors, Part 1.....	7
Creating and Visualizing Vectors in NumPy	7
Geometry of Vectors	10
Operations on Vectors	11
Adding Two Vectors	11
Geometry of Vector Addition and Subtraction	12
Vector-Scalar Multiplication	13
Scalar-Vector Addition	14
Transpose	15
Vector Broadcasting in Python	16
Vector Magnitude and Unit Vectors	17
The Vector Dot Product	18
The Dot Product Is Distributive	20
Geometry of the Dot Product	21

Other Vector Multiplications	22
Hadamard Multiplication	22
Outer Product	23
Cross and Triple Products	24
Orthogonal Vector Decomposition	24
Summary	28
Code Exercises	29
3. Vectors, Part 2.....	33
Vector Sets	33
Linear Weighted Combination	34
Linear Independence	35
The Math of Linear Independence	37
Independence and the Zeros Vector	38
Subspace and Span	38
Basis	41
Definition of Basis	44
Summary	46
Code Exercises	46
4. Vector Applications.....	49
Correlation and Cosine Similarity	49
Time Series Filtering and Feature Detection	52
<i>k</i> -Means Clustering	53
Code Exercises	57
Correlation Exercises	57
Filtering and Feature Detection Exercises	58
<i>k</i> -Means Exercises	60
5. Matrices, Part 1.....	61
Creating and Visualizing Matrices in NumPy	61
Visualizing, Indexing, and Slicing Matrices	61
Special Matrices	63
Matrix Math: Addition, Scalar Multiplication, Hadamard Multiplication	65
Addition and Subtraction	65
“Shifting” a Matrix	66
Scalar and Hadamard Multiplications	67
Standard Matrix Multiplication	67
Rules for Matrix Multiplication Validity	68
Matrix Multiplication	69
Matrix-Vector Multiplication	70
Matrix Operations: Transpose	72

Dot and Outer Product Notation	73
Matrix Operations: LIVE EVIL (Order of Operations)	73
Symmetric Matrices	74
Creating Symmetric Matrices from Nonsymmetric Matrices	74
Summary	75
Code Exercises	76
6. Matrices, Part 2.....	81
Matrix Norms	82
Matrix Trace and Frobenius Norm	83
Matrix Spaces (Column, Row, Nulls)	84
Column Space	84
Row Space	88
Null Spaces	88
Rank	91
Ranks of Special Matrices	94
Rank of Added and Multiplied Matrices	96
Rank of Shifted Matrices	97
Theory and Practice	98
Rank Applications	99
In the Column Space?	99
Linear Independence of a Vector Set	100
Determinant	101
Computing the Determinant	102
Determinant with Linear Dependencies	103
The Characteristic Polynomial	104
Summary	106
Code Exercises	107
7. Matrix Applications.....	113
Multivariate Data Covariance Matrices	113
Geometric Transformations via Matrix-Vector Multiplication	116
Image Feature Detection	120
Summary	124
Code Exercises	124
Covariance and Correlation Matrices Exercises	124
Geometric Transformations Exercises	126
Image Feature Detection Exercises	127
8. Matrix Inverse.....	129
The Matrix Inverse	129
Types of Inverses and Conditions for Invertibility	130

Computing the Inverse	131
Inverse of a 2×2 Matrix	131
Inverse of a Diagonal Matrix	133
Inverting Any Square Full-Rank Matrix	134
One-Sided Inverses	136
The Inverse Is Unique	138
Moore-Penrose Pseudoinverse	138
Numerical Stability of the Inverse	139
Geometric Interpretation of the Inverse	141
Summary	142
Code Exercises	143
9. Orthogonal Matrices and QR Decomposition.....	147
Orthogonal Matrices	147
Gram-Schmidt	149
QR Decomposition	150
Sizes of Q and R	151
QR and Inverses	154
Summary	154
Code Exercises	155
10. Row Reduction and LU Decomposition.....	159
Systems of Equations	159
Converting Equations into Matrices	160
Working with Matrix Equations	161
Row Reduction	163
Gaussian Elimination	165
Gauss-Jordan Elimination	166
Matrix Inverse via Gauss-Jordan Elimination	167
LU Decomposition	169
Row Swaps via Permutation Matrices	170
Summary	171
Code Exercises	172
11. General Linear Models and Least Squares.....	175
General Linear Models	176
Terminology	176
Setting Up a General Linear Model	176
Solving GLMs	178
Is the Solution Exact?	179
A Geometric Perspective on Least Squares	180
Why Does Least Squares Work?	181

GLM in a Simple Example	183
Least Squares via QR	187
Summary	188
Code Exercises	188
12. Least Squares Applications.....	193
Predicting Bike Rentals Based on Weather	193
Regression Table Using statsmodels	198
Multicollinearity	199
Regularization	199
Polynomial Regression	200
Grid Search to Find Model Parameters	204
Summary	206
Code Exercises	206
Bike Rental Exercises	206
Multicollinearity Exercise	207
Regularization Exercise	208
Polynomial Regression Exercise	210
Grid Search Exercises	210
13. Eigendecomposition.....	213
Interpretations of Eigenvalues and Eigenvectors	214
Geometry	214
Statistics (Principal Components Analysis)	215
Noise Reduction	216
Dimension Reduction (Data Compression)	217
Finding Eigenvalues	217
Finding Eigenvectors	220
Sign and Scale Indeterminacy of Eigenvectors	221
Diagonalizing a Square Matrix	222
The Special Awesomeness of Symmetric Matrices	224
Orthogonal Eigenvectors	224
Real-Valued Eigenvalues	226
Eigendecomposition of Singular Matrices	227
Quadratic Form, Definiteness, and Eigenvalues	228
The Quadratic Form of a Matrix	228
Definiteness	230
$\mathbf{A}^T \mathbf{A}$ Is Positive (Semi)definite	231
Generalized Eigendecomposition	232
Summary	233
Code Exercises	234

14. Singular Value Decomposition.....	241
The Big Picture of the SVD	241
Singular Values and Matrix Rank	243
SVD in Python	243
SVD and Rank-1 “Layers” of a Matrix	244
SVD from EIG	246
SVD of $\mathbf{A}^T \mathbf{A}$	247
Converting Singular Values to Variance, Explained	247
Condition Number	248
SVD and the MP Pseudoinverse	249
Summary	250
Code Exercises	251
15. Eigendecomposition and SVD Applications.....	255
PCA Using Eigendecomposition and SVD	255
The Math of PCA	256
The Steps to Perform a PCA	259
PCA via SVD	259
Linear Discriminant Analysis	260
Low-Rank Approximations via SVD	262
SVD for Denoising	263
Summary	263
Exercises	264
PCA	264
Linear Discriminant Analyses	269
SVD for Low-Rank Approximations	272
SVD for Image Denoising	275
16. Python Tutorial.....	279
Why Python, and What Are the Alternatives?	279
IDEs (Interactive Development Environments)	280
Using Python Locally and Online	280
Working with Code Files in Google Colab	281
Variables	282
Data Types	283
Indexing	284
Functions	285
Methods as Functions	286
Writing Your Own Functions	287
Libraries	288
NumPy	289
Indexing and Slicing in NumPy	289

Visualization	290
Translating Formulas to Code	293
Print Formatting and F-Strings	296
Control Flow	297
Comparators	297
If Statements	297
For Loops	299
Nested Control Statements	300
Measuring Computation Time	301
Getting Help and Learning More	301
What to Do When Things Go Awry	301
Summary	302
Index.....	303

